

Company Database (the provided database) – JOIN Queries

1. Display the department ID, department name, manager ID, and the full name of the manager.

The screenshot shows a SQL query window titled "SQLQuery1.sql - EB...BTESAM\ebtes (73))". The query is as follows:

```
select * from Employee;

-- SELECT
--      Dnum          AS DepartmentID,
--      Dname          AS DepartmentName,
--      EmpSSN         AS ManagerID,
--      e.FirstName + ' ' + e.LastName AS ManagerFullName
-- FROM Department808 d JOIN Employee e
--      ON d.EmpSSN = e.SSN;
```

The results pane shows a table with 5 rows and 5 columns: DepartmentID, DepartmentName, ManagerID, and ManagerFullName. The data is as follows:

	DepartmentID	DepartmentName	ManagerID	ManagerFullName
1	1	Human Resources	1	Ahmed Ali
2	2	Finance	2	Omar Hassan
3	3	Information Technology	3	Fatima Salim
4	4	Marketing	4	Yusuf Khalid
5	5	Operations	5	Layla Mahmoud

2. Display the names of departments and the names of the projects they control.

The screenshot shows a SQL query window titled "SQLQuery1.sql - EB...BTESAM\ebtes (73))". The query is as follows:

```
SELECT
    d.Dname AS DepartmentName,
    p.Pname AS ProjectName
FROM Department808 d JOIN Project p
    ON d.Dnum = p.Dnum;
```

The results pane shows a table with 5 rows and 2 columns: DepartmentName and ProjectName. The data is as follows:

	DepartmentName	ProjectName
1	Finance	Payroll System
2	Information Technology	Website Upgrade
3	Human Resources	Recruitment Drive
4	Marketing	Marketing Campaign
5	Operations	Logistics Optimization

3. Display full data of all dependents, along with the full name of the employee they depend on.

SQLQuery1.sql - EB...BTESAM\ebtes (73))*

```

----- JOIN-----

SELECT
    d.DepID,
    d.EmpSSN,
    d.Dnum,
    d.Gender,
    d.DBirthdate,
    e.FirstName + ' ' + e.LastName AS EmployeeFullName
FROM EMPDependent d JOIN Employee e
ON d.EmpSSN = e.SSN;

```

100 %

Results Messages

	DepID	EmpSSN	Dnum	Gender	DBirthdate	EmployeeFullName
1	1	1	1	0	2010-06-15	Ahmed Ali
2	2	2	2	1	2012-11-03	Omar Hassan
3	3	3	3	0	2015-09-25	Fatima Salim
4	4	4	4	1	2018-04-12	Yusuf Khalid
5	5	5	5	0	2016-07-30	Layla Mahmoud

4. Display the project ID, name, and location of all projects located in Cairo or Alex.

```

SELECT
    Pnum AS ProjectID,
    Pname AS ProjectName,
    PLocation AS ProjectLocation
FROM Project
WHERE City IN ('Muscat', 'Muscat');

```

100 %

Results Messages

	ProjectID	ProjectName	ProjectLocation
1	1	Payroll System	Main Office
2	4	Marketing Campaign	Main Office

5. Display all project data where the project name starts with the letter 'A'.

```
SELECT *
FROM Project
WHERE Pname LIKE 'A%';
```

100 %

Results Messages

	Pnum	Pname	PLocation	City	Dnum
1	4	Marketing Campaign	Main Office	Muscat	4

6. Display the IDs and names of employees in department 30 with a salary between 1000 and 2000 LE.

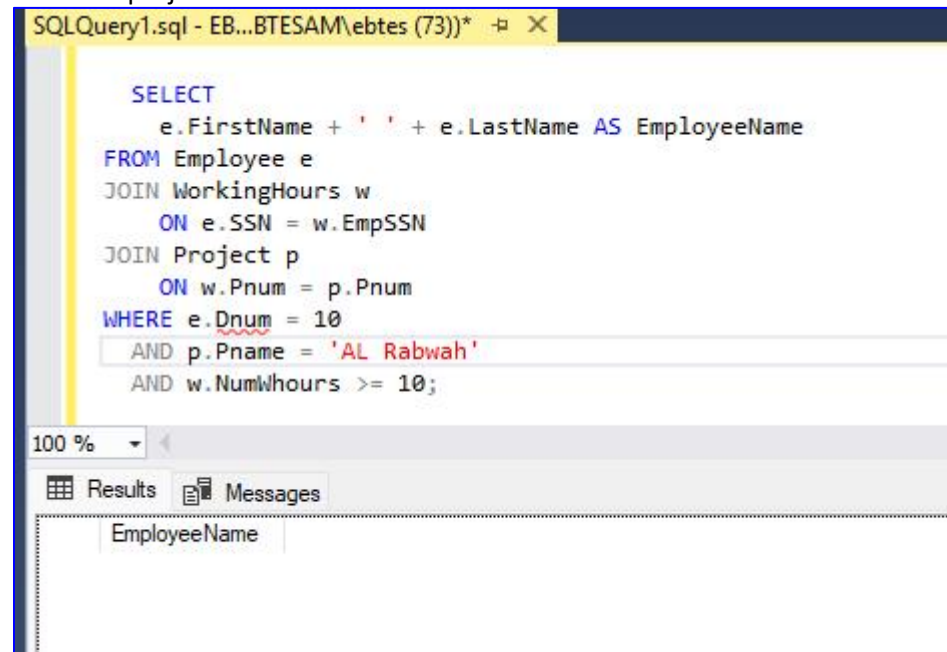
```
SELECT
    e.SSN AS EmployeeID,
    e.FirstName + ' ' + e.LastName AS EmployeeName,
    d.Dname AS DepartmentName
FROM Employee e
JOIN Department808 d
    ON e.Dnum = d.Dnum;
```

100 %

Results Messages

	EmployeeID	EmployeeName
1	1	Ahmed Ali
2	2	Omar Hassan
3	3	Fatima Salim
4	4	Yusuf Khalid
5	5	Layla Mahmoud

7. Retrieve the names of employees in department 10 who work ≥ 10 hours/week on the "AL Rabwah" project.

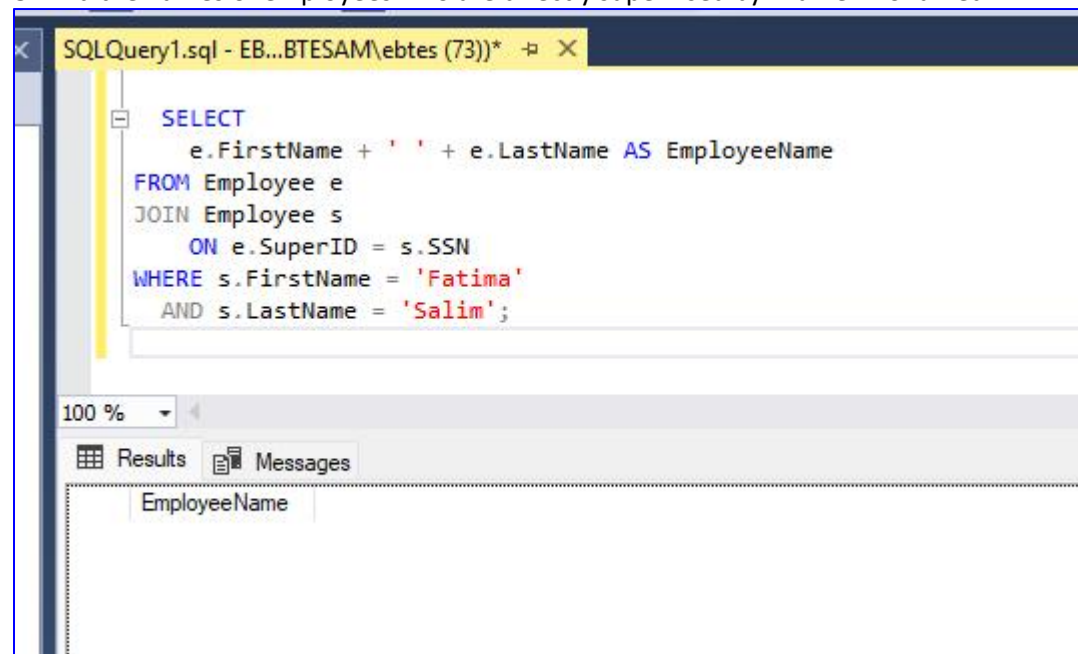


The screenshot shows a SQL query window titled "SQLQuery1.sql - EB...BTESAM\ebtes (73))". The query is as follows:

```
SELECT
    e.FirstName + ' ' + e.LastName AS EmployeeName
FROM Employee e
JOIN WorkingHours w
    ON e.SSN = w.EmpSSN
JOIN Project p
    ON w.Pnum = p.Pnum
WHERE e.Dnum = 10
    AND p.Pname = 'AL Rabwah'
    AND w.NumHours >= 10;
```

Below the query, the "Results" tab is active, showing a table with one column: "EmployeeName".

8. Find the names of employees who are directly supervised by "Kamel Mohamed".

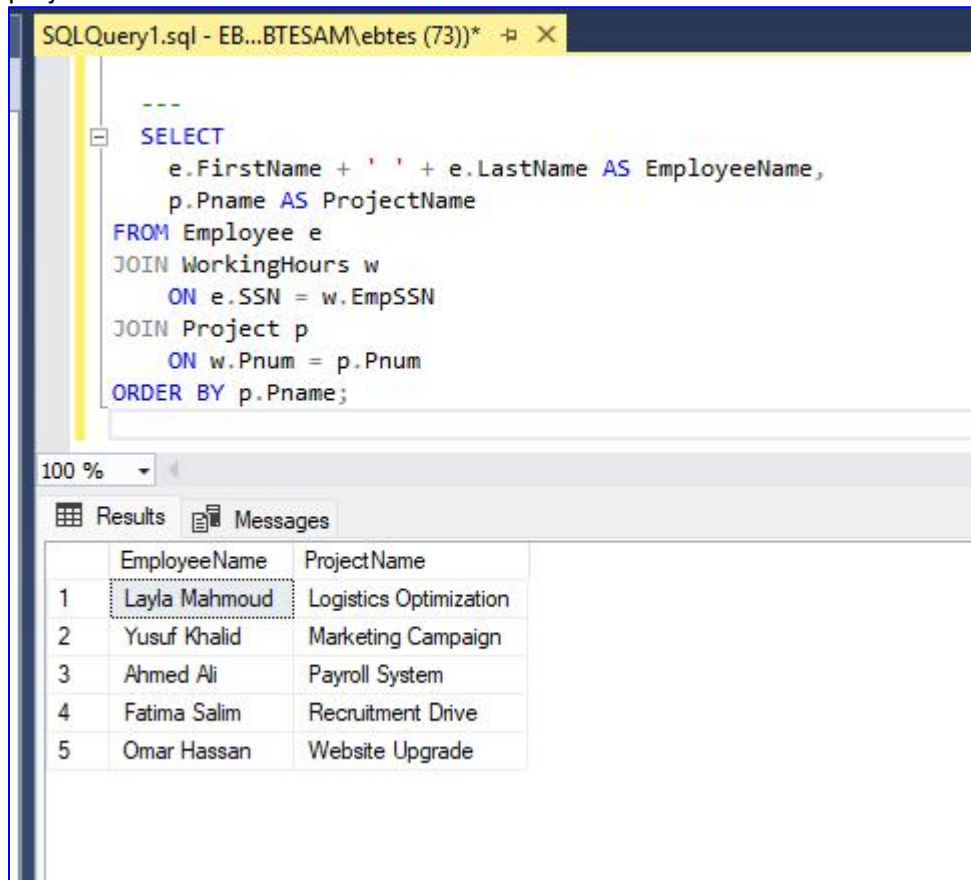


The screenshot shows a SQL query window titled "SQLQuery1.sql - EB...BTESAM\ebtes (73))". The query is as follows:

```
SELECT
    e.FirstName + ' ' + e.LastName AS EmployeeName
FROM Employee e
JOIN Employee s
    ON e.SuperID = s.SSN
WHERE s.FirstName = 'Fatima'
    AND s.LastName = 'Salim';
```

Below the query, the "Results" tab is active, showing a table with one column: "EmployeeName".

9. Retrieve the names of employees and the names of the projects they work on, sorted by project name.



The screenshot shows a SQL query window titled "SQLQuery1.sql - EB...BTESAM\ebtes (73))". The query is as follows:

```
---  
SELECT  
    e.FirstName + ' ' + e.LastName AS EmployeeName,  
    p.Pname AS ProjectName  
FROM Employee e  
JOIN WorkingHours w  
    ON e.SSN = w.EmpSSN  
JOIN Project p  
    ON w.Pnum = p.Pnum  
ORDER BY p.Pname;
```

Below the query window, the "Results" tab is active, displaying a table with 5 rows and 2 columns: "EmployeeName" and "ProjectName". The results are sorted by project name.

	EmployeeName	ProjectName
1	Layla Mahmoud	Logistics Optimization
2	Yusuf Khalid	Marketing Campaign
3	Ahmed Ali	Payroll System
4	Fatima Salim	Recruitment Drive
5	Omar Hassan	Website Upgrade

10. For each project located in Cairo, display the project number, controlling department name, manager's last name, address, and birthdate.

SQLQuery1.sql - EB...BTESAM\ebtes (73))*

```
---  
SELECT  
    p.Pnum AS ProjectNumber,  
    d.Dname AS DepartmentName,  
    m.LastName AS ManagerLastName,  
    m.SSN AS ManagerID,  
    m.Birthdate AS ManagerBirthdate  
FROM Project p  
JOIN Department808 d  
    ON p.Dnum = d.Dnum  
JOIN Employee m  
    ON d.EmpSSN = m.SSN  
WHERE p.City = 'Muscat';
```

100 %

Results Messages

	ProjectNumber	DepartmentName	ManagerLastName	ManagerID	ManagerBirthdate
1	1	Finance	Hassan	2	1990-09-23
2	4	Marketing	Khalid	4	1988-06-18

11. Display all data of managers in the company.

SQLQuery1.sql - EB...BTESAM\ebtes (73)) * -> X

```

SELECT e.*
FROM Employee e
WHERE e.SSN IN (
    SELECT EmpSSN
    FROM Department808
    WHERE EmpSSN IS NOT NULL
);

```

100 %

Results Messages

	SSN	FirstName	LastName	Birthdate	Gender	SuperID	Salary	Dnum
1	1	Ahmed	Ali	1985-04-12	1	NULL	1500.00	1
2	2	Omar	Hassan	1990-09-23	0	NULL	NULL	2
3	3	Fatima	Salim	1978-01-30	1	NULL	NULL	3
4	4	Yusuf	Khalid	1988-06-18	0	NULL	NULL	4
5	5	Layla	Mahmoud	1995-12-05	1	NULL	NULL	5

12. Display all employees and their dependents, even if some employees have no dependents.

SQLQuery1.sql - EB...BTESAM\ebtes (73)) * -> X

```

SELECT
    e.SSN AS EmployeeID,
    e.FirstName + ' ' + e.LastName AS EmployeeName,
    d.DepID AS DependentID,
    d.Gender AS DependentGender,
    d.DBirthdate AS DependentBirthdate
FROM Employee e
LEFT JOIN EMPDependent d
    ON e.SSN = d.EmpSSN
ORDER BY e.SSN;

```

100 %

Results Messages

	EmployeeID	EmployeeName	DependentID	DependentGender	DependentBirthdate
1	1	Ahmed Ali	1	0	2010-06-15
2	2	Omar Hassan	2	1	2012-11-03
3	3	Fatima Salim	3	0	2015-09-25
4	4	Yusuf Khalid	4	1	2018-04-12
5	5	Layla Mahmoud	5	0	2016-07-30